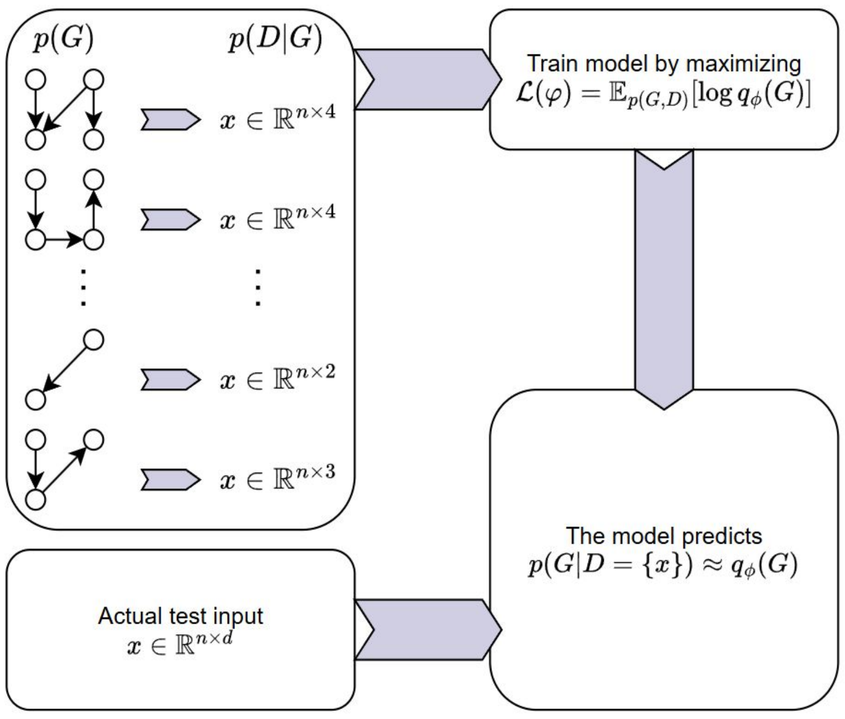
universität freiburg

Causal feature selection using AVICI

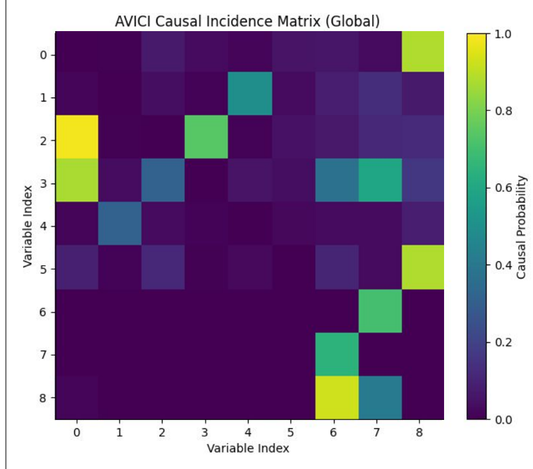
Hani Alnahas, Ole Ossen

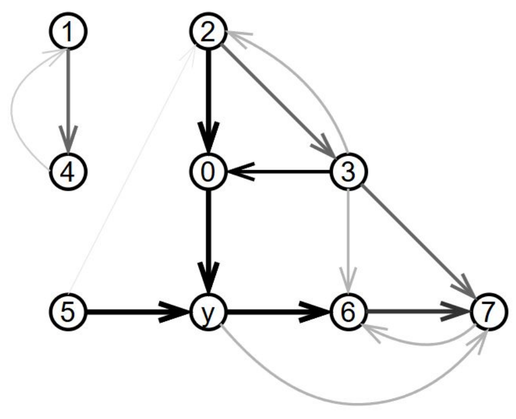
Reminder on AVICI [1]



Train a Transformer encoder with **8** Layers, each with **4** residual sublayers, to predict the underlying SCM of the input data. The model is permutation invariant across both the sample and the variable dimensions

DAGs





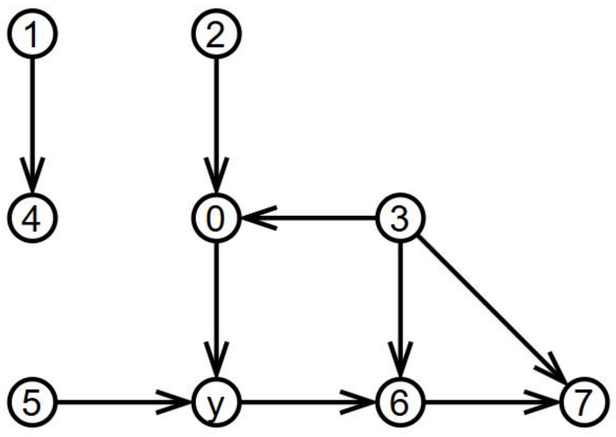
The output of AVICI on a given dataset is a probabilistic adjacency matrix of a directed acyclic graph (DAG).

Feature Selection

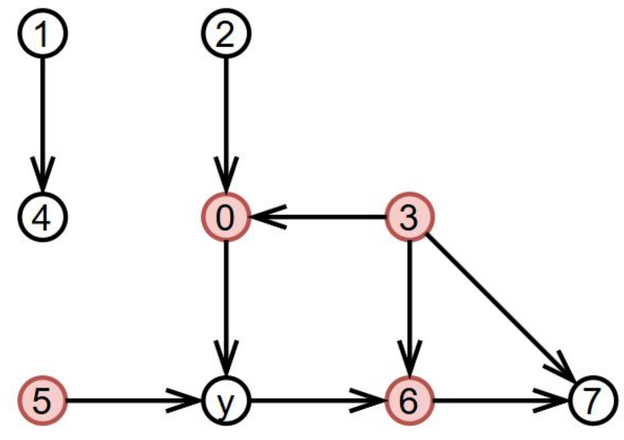
Feature selection: only use a subset of the features to predict target column.

Some advantages:

- Less memory required
- Faster training and inference
- Better interpretability
- Better performance



Question: Given data sampled from an SCM, which features should we use to predict *y*?



Answer: the parents, children, and co-parents of children of *y*. Then we have *conditional independence* [2, Section 1.2.3]:

$$\mathbb{P}(y \mid \text{red features}) = \mathbb{P}(y \mid \text{all features})$$

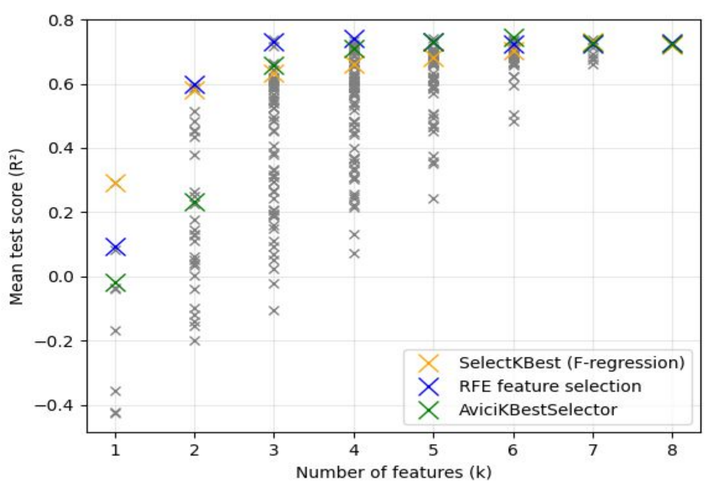
Experiments

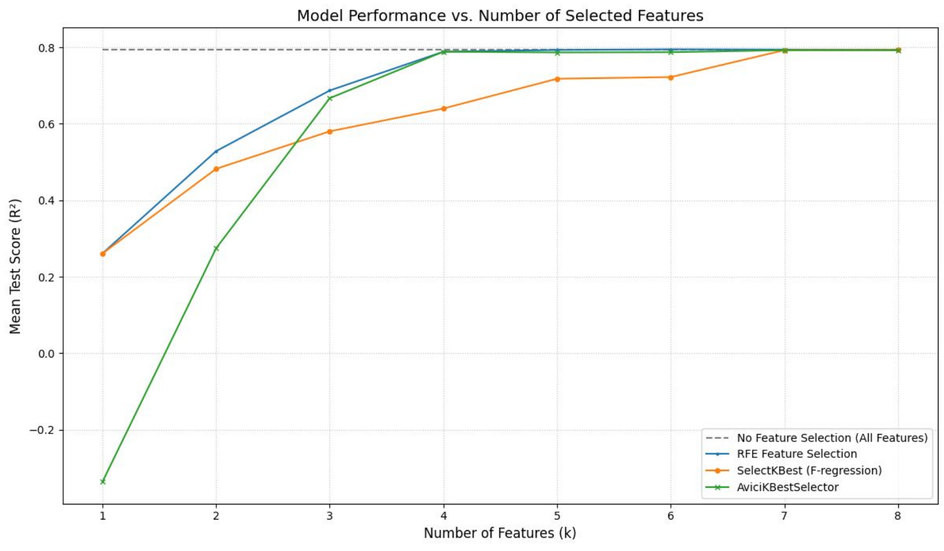
As a sample of AVICI's ability to perform feature selection, we tested it with a random forest regressor on the California Housing dataset. All results are averaged over 5-fold cross-validation.

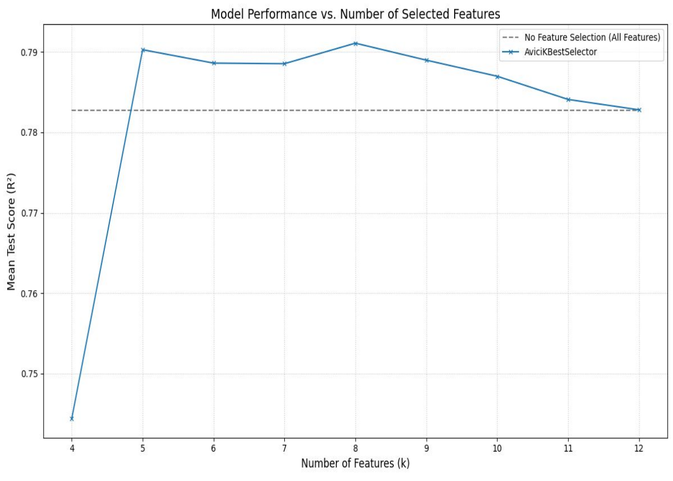
Bottom left: R^2-score as a function of the number of selected features, for three different feature selectors.

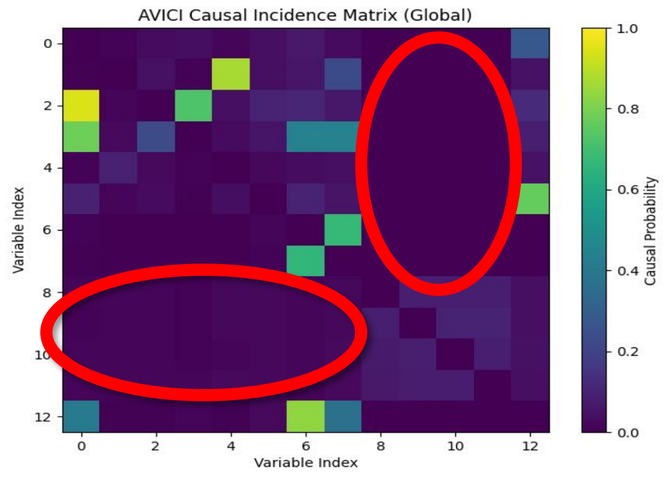
Top right: exhaustive search over all feature subsets.

Bottom right: four noise features are added to the dataset, all of which are dropped by AVICI for *k=8.*









[1] Lorch, Sussex, Rothfuss, Krause, Schölkopf. *Amortized Inference for Causal Structure Learning*, NeurIPS 2022

[2] Pearl. *Causality: Models, reasoning, and inference* (2nd ed.). Cambridge University Press, 2009